CF-MoA: Taxonomy-Guided Benchmarking and Agent Orchestration for Counterfactual Medical Reasoning

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Abstract

Medical question answering requires selective responses to changing evidence: a new finding can justify an additional treatment, invalidate an earlier recommendation, reorder competing diagnoses, or leave a conclusion unchanged. We introduce a five-relation functional taxonomy that connects these behaviors across heterogeneous tasks and guides the design of specialized reasoning operations. From 11 sources, we construct a benchmark of 6,104 units and 13,000 target judgments, preserving native task interfaces and source-specific scoring. Evaluating seven medical question-answering configurations reveals recurring weaknesses in support updating and candidate comparison that aggregate accuracy obscures. We address these weaknesses with a counterfactual mixture of agents (CF-MoA): five functional roles for support addition, support removal, candidate comparison, model-based consequence calculation, and stable readout, organized by a sparse controller. On this development benchmark, the corresponding heads improve all 21 comparisons across seven configurations and three support-update and candidate-comparison categories. The complete controller raises the aggregate score of two MedRAG configurations from 48.20 to 58.29 and from 47.35 to 59.36, with gains on both answer-changing and answer-preserving pairs. Component controls identify explicit support execution and catalog-based re-decision as useful mechanisms. The taxonomy, benchmark, and controller connect behavioral diagnosis to concrete operations for improving counterfactual medical reasoning.

1 Introduction

A medical answer depends on the evidence that supports it. When a patient's condition changes, a system may need to add a newly indicated treatment, withdraw an option whose prerequisites no longer hold, or reconsider the leading diagnosis. Other changes should leave the answer intact. Selective updating requires tracking how the changed information bears on the particular judgment being requested.

Consider a treatment supported by two independent indications. Removing one indication leaves the other available to justify treatment; withdrawing the treatment requires checking both support paths. A diagnostic update requires a different operation: new evidence can strengthen a candidate while another diagnosis remains more likely. The final answer's change or preservation captures the outcome of these decisions. The underlying operation—adding support, removing support, or comparing alternatives—explains what the system must compute.

Retrieval-augmented generation (RAG) gives medical language models access to external knowledge, and agent-based systems add iterative retrieval, multiple answers, or explicit control of reasoning (Xiong et al., 2024; Li et al., 2026; Wu et al., 2026; Jiang et al., 2025). Using that evidence requires an appropriate decision procedure for the requested update. Existing counterfactual, perturbation, and robustness resources test complementary aspects of this ability through different units, answer spaces, and evaluation rules. A shared behavioral vocabulary can connect their observations while preserving their original tasks.

We introduce a taxonomy of five functional relations between a change and a target judgment: support addition, support removal, candidate comparison, consequence calculation, and target preservation. The categories can overlap: the same change may add evidence for a candidate and require a new comparison among alternatives. We use this taxonomy to organize an inventory of 26 candidate resources and construct a benchmark spanning 11 sources. Its 6,104 evaluation units retain the native questions, options, target definitions, and source-specific scoring definitions.

Functional evaluation reveals weaknesses hidden by aggregate rankings. Seven configurations from six system families achieve aggregate scores between 47.35 and 51.61, while their scores on support removal range from 7.59 to 41.77. Different systems lead on different functional categories. These profiles motivate an operation-centered approach: identify the computation required by the observed failure mode, then attach a component that performs it through the existing interface.

We organize these components as a counterfactual mixture of agents (CF-MoA), with five functional roles. Its support agents execute three-valued rules, its comparison agent makes an explicit decision over a diagnosis catalog, its consequence agent applies local interventions to a supplied executable model, and its preservation agent stabilizes native answers through structured readout. A sparse controller selects the applicable operation from the visible input and available capabilities. This design gives each agent a concrete responsibility and allows compatible operations to share an implementation.

Our contributions are threefold. First, we introduce a functional taxonomy and a provenance-preserving benchmark that connect diverse medical counterfactual and robustness tasks. Second, we provide a seven-configuration diagnostic evaluation, together with head-level comparisons that improve all 21 system–category combinations on support addition, support removal, and candidate comparison. Third, we develop CF-MoA and evaluate the complete adopted controller on two MedRAG configurations, obtaining aggregate gains of 10.09 and 12.01 percentage points and improved correctness on both changing and preserving pairs. The accompanying controls explain which operations account for these improvements and when further aggregation adds little value.

2 Related Work

Behavioral evaluation and counterfactual tests. Behavioral testing organizes evaluation around capabilities and expected responses to controlled changes. CheckList combines linguistic capabilities with minimum functionality, invariance, and directional tests (Ribeiro et al., 2020). Contrast sets probe local decision boundaries through small, meaningful edits that typically change the reference label (Gardner et al., 2020). Dynabench develops this diagnostic perspective through human-and-model-in-the-loop data collection (Kiela et al., 2021). Our five-category taxonomy applies this behavioral perspective to medical tasks by recording the changed condition, the requested decision, and the evidence needed to justify revision or preservation. Overlapping categories represent cases that require several operations, giving tests with different native formats a shared functional interpretation.

Medical counterfactual and robustness benchmarks. Medical datasets expose several distinct consequences of perturbing an input. MedEinst changes discriminative evidence to test differential diagnosis (Chen et al., 2026), while MedPIC-Bench provides patient-specific medication questions with counterfactual task families (MedPIC-Bench). HIV treatment scenarios examine causal consistency across interventions, interactions, and side effects (Prosperi et al., 2026). Counterfactual Patient Variations and DiversityMedQA examine demographic sensitivity in clinical decisions (Benkirane et al., 2025; Rawat et al., 2024); cultural-cue perturbations and AMQA extend this line to cultural context and adversarial medical questions (Rezaei and Shakeri, 2026; Xiao et al., 2025). MedPerturb varies gender cues, language style, and presentation format (Gourabathina et al., 2025). MedCounterFact studies how models follow supplied medical evidence when its content conflicts with plausibility or safety (Mo et al., 2026). MedRGB and BioRAB assess retrieval conditions, including misleading or counterfactual evidence (Ngo et al., 2024; Li et al., 2025). We bring eligible instances from these eleven sources into one benchmark while preserving their native questions, answer spaces, and reference relations. The resulting functional labels connect heterogeneous perturbations to the reasoning operation that each instance actually tests.

Medical RAG. Retrieval-augmented generation combines a language model with external information (Lewis et al., 2020). MIRAGE and the MedRAG toolkit examine how corpora, retrievers, and language models affect medical question answering (Xiong et al., 2024). MedRGAG connects retrieved and parametric knowledge (Li et al., 2026); MA-RAG uses multiple rounds of agentic retrieval and reasoning (Wu et al., 2026). MedGENIE supplies generated contexts for medical questions (Frisoni et al., 2024), and TC-RAG maintains a memory stack to control iterative retrieval (Jiang et al., 2025). Our seven-configuration evaluation diagnoses how these approaches to acquiring and organizing knowledge perform on distinct functional requirements. Operation-specific heads then use their outputs and admissible evidence to recompute support or compare diagnostic candidates, connecting system-level diagnosis to specific decision operations.

Agent aggregation. Self-consistency aggregates answers across sampled reasoning paths (Wang et al., 2023), while Mixture-of-Agents passes multiple agents' outputs between successive layers (Wang et al., 2025). Contextual calibration addresses prompt-dependent answer preferences (Zhao et al., 2021). These approaches motivate our controls for additional sampling, candidate comparison, and aggregation. MedGuideX transforms executable guideline logic into factual and counterfactual supervision for post-training medical models (Shen et al., 2026); our support agents execute guideline logic during inference to recompute a current answer. CF-MoA assigns five functional responsibilities and coordinates their execution through a shared input contract and routing interface. The evaluation measures both the gains from integrating specialized operations and the incremental effects of candidate verification, tying each system change to its intended behavior.

3 A Functional Benchmark for Medical Counterfactual Reasoning

Our benchmark evaluates the relationship between a change and a specified medical judgment. A *comparison unit* links a source-provided variant to its reference or identifies an explicit within-question hypothetical. A *target judgment* identifies what is being assessed within

that unit. Separating these levels allows several functional annotations within one clinical narrative while preserving the source's original question and answer contract. The taxonomy covers condition changes, hypothetical evidence, and perturbations evaluated through native task performance.

3.1 Five overlapping functional relations

R1: adding support. A change introduces identifiable evidence for a target judgment. For example, newly reported wheezing can support a bronchitis candidate without establishing it as the unique diagnosis. **R2: withdrawing support.** An identifiable basis for a judgment becomes invalid; the annotation states which basis is lost and preserves any independent support. **R3: comparing alternatives.** A change requires reassessing the relative support for at least two identifiable candidates. Replacing a history of spontaneous pneumothorax with deep-vein thrombosis illustrates a change relevant to the comparison between pneumothorax and pulmonary embolism in MedEinst Chen et al. (2026). **R4: deriving consequences.** A changed premise requires propagating an established relationship to an outcome not directly supplied. **R5: preserving a judgment.** A real change leaves the evidence relevant to a specified target intact; the annotation gives a reason for that invariance.

The taxonomy captures overlapping functions at the target level. The same observation may add support to one diagnosis and alter its comparison with another. Each annotation records the change, target, applicable relations, supporting text, and conclusions that cannot be inferred; the parent unit receives the union of its target labels. Omission of a symptom does not establish its explicit absence, and an unchanged answer alone does not establish invariance. Native comparison axes are preserved: an internal hypothetical and an externally edited patient attribute receive separate treatment when they address different judgments.

3.2 Collection, qualification, and sampling

We registered 26 candidate sources and obtained 15, comprising 14 text resources and one multimodal metadata release lacking required images. Their normalized archive contains 55,991 units. Source acquisition, semantic classification, and scoring eligibility are recorded separately. Qualification requires the source materials needed to answer the native task and an established scoring interface. This process yielded an eligible pool of 38,123 units and 77,781 target judgments from 11 sources. The included resources cover diagnostic and prescribing edits Chen et al. (2026); MedPIC-Bench, treatment hypotheticals Prosperi et al. (2026), demographic and cultural perturbations Benkirane et al. (2025); Rezaei and Shakeri (2026); Xiao et al. (2025); Rawat et al. (2024), clinical-service judgments Gourabathina et al. (2025), hypothetical research evidence Mo et al. (2026), and evidence robustness Ngo et al. (2024); Li et al. (2025).

Language-model agents performed structured semantic annotation and review, supported by cross-checks and targeted audits. Review assesses both the functional relation and the reference answer's suitability for scoring. A valid invariance annotation, for example, remains in the archive even when its unreliable reference is excluded from scoring. Targeted review corrected 172 judgments across 130 units, with the earlier versions retained. The pipeline records unit identifiers, source locations, eligibility decisions, target-to-input mappings, and correction history, making the final evaluation traceable to the acquired materials.

The frozen v5 evaluation retains all eligible R1–R4 targets and samples 450 R5 units without replacement from each of eight sources. Sampling uses sorted source-specific identifiers and a fixed seeded procedure, without consulting model predictions. Original semantic labels remain available, while a separate evaluation-label field specifies membership in this evaluation. This yields 6,104 unique comparison units, with category counts of 2,866/79/733/13/3,600 for R1–R5. Category memberships overlap; in particular, all 733 R3 units also belong to R1. Equal R5 allocations preserve source coverage while reducing the much larger invariance pool.

3.3 Native inputs and scoring

The selected units contain 13,000 target judgments and materialize into 13,905 distinct model inputs, including reference endpoints. Each method has 15,616 native scoring mappings: one input may support multiple targets, and shared references can serve several units. Models receive the current question, native options, required evidence, and output contract. Reference and variant endpoints are answered independently; gold answers, functional labels, annotation rationales, and the opposite endpoint's answer remain outside inference.

Native interfaces retain each source's decision task. MedEinst uses open diagnosis, MedPIC uses multiple-answer selection, and MedPerturb expands eligible visit, management, and resource judgments separately. Hypothetical research evidence remains attached to its relation question. MedRGB retains the locally frozen six evidence conditions, with the fully relevant condition serving as the reference endpoint. Systems with retrieval permission can additionally retrieve external evidence, combining fixed perturbed evidence with dynamic retrieval.

For unit u, let $\mathcal{J}_u$ contain its eligible non-reference native scoring mappings and let $c_j \in \{0, 1\}$ denote primary

correctness. We report

$$s_u = \frac{1}{|\mathcal{J}_u|} \sum_{j \in \mathcal{J}_u} c_j, \qquad \text{ALL} = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} s_u, \qquad (1)$$

where $|\mathcal{U}| = 6{,}104$. Category scores apply the same aggregation to their eligible members and mappings. Tables report these scores as percentages. Thus, ALL weights comparison units equally, rather than averaging category percentages or pooling all model inputs.

Scoring retains native choice keys, multi-answer sets, relation labels, and frozen diagnosis normalization. Invalid outputs and unresolved protocol failures stay in the denominator. We additionally distinguish variant correctness, answer agreement, and both-endpoints-correct performance. The last measure requires successful answers on both sides, making it useful for evaluating whether systems respond to relevant changes while preserving correct judgments under irrelevant ones. These functional strata provide the diagnostic basis for the specialized operations introduced next.

4 CF-MoA: Sparse Orchestration of Five Functional Roles

CF-MoA implements five functional agent roles: adding support (A1), withdrawing support (A2), comparing candidates (A3), deriving consequences (A4), and preserving task-relevant judgments (A5). These roles correspond to R1–R5 and operate through a sparse controller. The adopted configuration, denoted B, selects an applicable operation from the current input's capabilities and returns a native answer (Figure 1). The base system supplies its current answer and actual inference context; agents share backbone weights and, where applicable, deterministic components.

4.1 Capability routing and native readout

The controller selects operations by task format and available resources. Its fixed priority routes diagnosis-format inputs to A3, eligible rule tasks to the shared A1/A2 kernel, inputs providing a complete executable model to A4, and supported single-choice, robustness, or relation tasks with saved original messages to A5. Other inputs retain the base response. Routing uses visible options and question structure alongside format and resource availability; source identifiers, benchmark relation labels, gold answers, and paired endpoints are excluded. Capability-based routing allows a component developed on one functional subset to operate on compatible inputs elsewhere in the benchmark.

Each role operates within a defined domain: finite prescribing rules for A1/A2, a fixed diagnosis catalog for A3, a supplied executable mechanism and state for A4, and supported native decision formats for A5. In the complete v5 evaluation, each backbone routes 3,560 inputs to A3, 123 to A1/A2, 10,178 to A5, and 44 to passthrough. A4 has no qualifying clinical v5 input; its supplied-model evaluation is reported separately. The controller records the selected route, requests, operation traces, and complete returned response.

4.2 A1/A2: scoped support recomputation

A1 and A2 recompute support with a shared kernel of 71 manually encoded rules drawn from selected recommendations in the 2023 American Geriatrics Society Beers Criteria (2023 American Geriatrics Society Beers Criteria Update Expert Panel, 2023). Eligibility checks require a supported multiple-choice format, an age of at least 65 in the implemented question pattern, and a recognized medication option. The kernel identifies the requested action and scope, such as avoidance or renal dose adjustment. One call to the original backbone extracts the required patient facts and evidence sentence identifiers from the numbered current question. The extraction input excludes the base answer and retrieved documents. Missing or rejected evidence produces an unknown value.

The executor evaluates conditions, exceptions, and independent support paths using MET, CONTRADICTED, and UNKNOWN. Conjunction combines requirements within a path; disjunction preserves independent paths. Consequently, invalidating one risk does not remove an option supported by another satisfied risk. Let S_0 be the initial valid option set and let M, C, U partition options by their aggregate status. A1 adds $M \setminus S_0$; A2 removes $C \cap S_0$. For ordinary valid sets, the result is

$$S^\star = M \cup (S_0 \cap U). \qquad (2)$$

The native assembler additionally handles exclusive "None" options, invalid initial answers, and empty outputs. Both roles therefore use one fact extraction and one execution, while retaining separate addition and withdrawal traces.

4.3 A3: catalog-based candidate comparison

A3 compares all 49 disease names in the public DDXPlus catalog Fansi Tchango et al. (2022). The same backbone reads the current patient and the base system's actual context, with diagnoses bound to scoreable codes under six fixed catalog orders. For each order, it subtracts a content-free log-score prior from the patient-conditioned code score, following the general calibration principle of separating contextual evidence from answer preferences Zhao et al. (2021). The prior is the logarithm of the mean of three content-free candidate distributions,

computed separately for each order; its subtraction coefficient is fixed at one. A3 selects the highest corrected score per order and votes across orders, breaking vote ties using average corrected probabilities. The selected catalog entry is mapped back to the native diagnosis. The catalog supplies candidates for this decision stage; the scorer uses its separate, frozen normalization dictionary.

4.4 A4: executing a local intervention

A4 compiles an intervention into a typed local patch and executes it to compute its consequences. Its input supplies the executable model, parameters, complete initial state, and factual action schedule. The patch preserves all unmodified actions. The numerical executor runs factual and patched trajectories from the same starting state and projects their consequences onto the requested time and outcome. Our implementation uses the public glucose–insulin simulator Xie (2025). Local patches reduce the need to regenerate a complete action plan, allowing intervention fidelity to be evaluated against a same-tool full-plan control. The returned consequence follows from the supplied mechanism and executed actions.

4.5 A5: task-adapted robust readout

A5 implements task-adapted robust readout through component F. For ordinary supported tasks, F preserves the complete original messages and samples up to five native responses at temperature 0.7, seeds 42–46, and a 2,048-token output limit. It votes over valid answer keys, stops once one key receives three votes, and returns the earliest complete response carrying a winning key. Final vote ties use earliest occurrence; if no vote is valid, the first response remains invalid. This adapts sampling and answer aggregation to the native output contract Wang et al. (2023).

For explicit premise–hypothesis tasks with two polarity options, F uses a natural language inference (NLI) branch. It scores both A/B label mappings directly, generates one current-question analysis, and scores both mappings again conditioned on that analysis. Each two-label distribution is normalized before the four semantic distributions are averaged. The highest mean determines the answer; exact ties follow native option order. A final model call generates the complete native response with its answer field constrained to that decision. This path uses four one-token score callbacks and two complete generations at temperature zero, preserving the original evidence and required auxiliary fields. Missing code scores require a matching score request and remain visible in the cost ledger.

5 Experimental Setup

Seven system configurations. We evaluate direct answering with Llama-3.1-8B-Instruct (Dubey et al., 2024); an all-Llama implementation of MedRGAG; intrinsic MA-RAG with Qwen3-8B (Yang et al., 2025); MedRAG with each of these two backbones; MedGENIE with Llama; and TC-RAG with Qwen. These are seven configurations from six system families. We refer to the two MedRAG configurations as MedRAG-Llama and MedRAG-Qwen. The selected implementations use structured output handling for MedRAG-Llama, the second local prompt version of MedGENIE, and typed control for TC-RAG. MA-RAG uses eight solvers per round, up to four rounds, with early agreement stopping. Its available retrieval collection contains PubMed, textbooks, and Wikipedia; the other retrieval configurations use their own corpus and retrieval settings. This comparison evaluates complete systems with their respective retrieval and inference budgets. Appendix D records the implementation identities and reuse accounting.

Head and system evaluations. The evaluation tests head transfer across seven configurations and complete controller performance on two MedRAG backbones. The head evaluation attaches the support-update and comparison operations to the saved contexts and original answers of all seven configurations. It covers all 2,866 support-addition units, 79 support-removal units, and 733 candidate-comparison units, with overlapping category membership as specified in Section 3. Diagnosis heads use the corresponding system contexts; the rule branches reuse two backbone-specific sets of extracted facts and combine them offline with each system's original answer. The full-system evaluation executes the adopted CF-MoA controller, denoted B, on MedRAG-Llama and MedRAG-Qwen. Each configuration covers the same 13,905 unique inputs and 15,616 native scoring mappings. Head and full-system results are reported separately.

Mechanism controls. The mechanism controls examine the contributions of candidate presentation, support operations, local intervention, and answer aggregation. On the 680-unit diagnostic subset, we compare original answers, one catalog decision, six uncalibrated decisions, and calibrated decisions. A 123-input control removes the addition or removal operation while retaining the same extracted facts and rule statuses. Model-based consequence execution is evaluated separately on 24 virtual profiles with 12 questions each. For the full system, we evaluate three candidate-selection variants and an expanded sampling pool; their six-arm design and complete results appear in Appendix C.

Metrics and uncertainty. The primary metric is the native unit-macro score defined in Section 3. Category

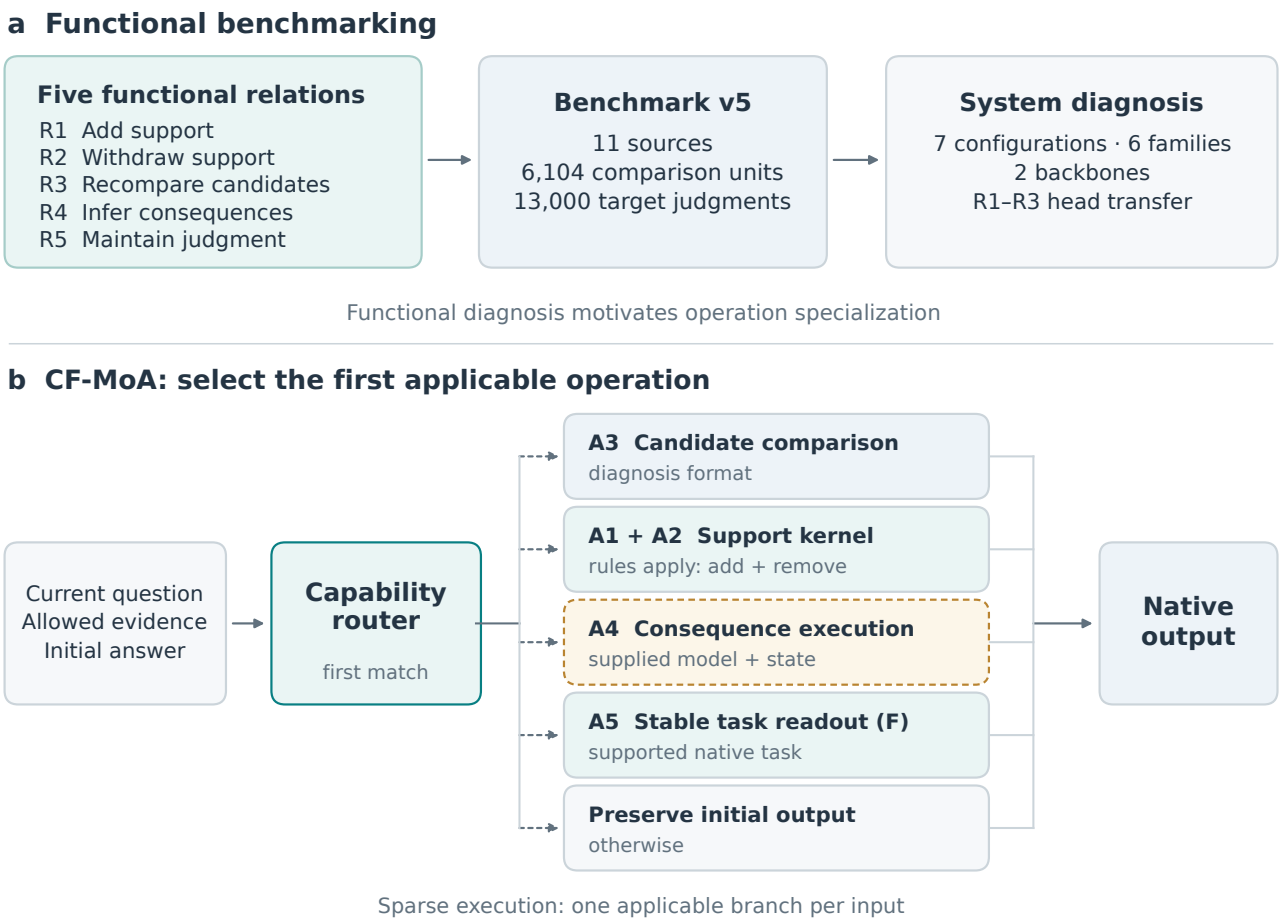


Figure 1: From functional diagnosis to agent orchestration. The benchmark describes five relations between a change and a target judgment. CF-MoA implements the corresponding operations through a sparse controller: support addition and removal share a rule executor, candidate comparison uses a diagnosis catalog, consequence calculation requires an executable model, and stable readout handles the remaining eligible interfaces. Routing uses the current input and available capabilities; A4 is evaluated separately on supplied-model tasks.

scores use their respective member units. We additionally measure paired correctness: both the reference and changed input must be correct. We report this separately when the target answer should be maintained and when it should change, using 6,838 and 1,780 scored pairs, respectively. For paired system contrasts, we draw 5,000 bootstrap replicates using seed 240924, keeping related units together. The aggregate, preserving-pair, and changing-pair analyses contain 4,766, 2,806, and 1,780 parent families, respectively. The resulting 95% intervals describe variability within this benchmark, which was also used for method development.

Computation. The adopted components use the same 8B backbones as their host configurations. The full-system ordinary-answer sampler uses temperature 0.7, top-p = 1, and a 2,048-token output budget. Component-specific scoring and execution settings are recorded with the released configurations. Cost accounting reports measured component requests and tokens; original retrieval costs are unavailable. Invalid outputs remain in every scoring denominator.

6 Results

6.1 Functional evaluation exposes different weaknesses

The seven baselines occupy a narrow range on the aggregate metric but exhibit substantially different functional profiles (Table 1). MA-RAG has the highest baseline aggregate score, 51.61, and the highest preservation score, 73.37, yet scores 8.59 on candidate comparison. MedRAG-Qwen leads the baselines on support removal at 41.77, while MedRAG-Llama leads on candidate comparison at 30.42. Direct answering reaches 48.10 overall but only 8.86 on support removal. Thus, a system can answer many ordinary or preservation-oriented items

Table 1: Complete v5 results for seven baseline configurations and two CF-MoA instantiations. Scores are percentages of native-unit means; bold denotes the highest value in each column.

System	Backbone	ALL	R1	R2	R3	R4	R5
Comparison units		6,104	2,866	79	733	13	3,600
Direct	Llama	48.10	30.04	8.86	17.87	38.46	65.70
MedRGAG	Llama	48.81	31.33	7.59	24.69	53.85	65.88
MA-RAG	Qwen	51.61	26.73	34.18	8.59	38.46	**73.37**
MedRAG	Llama	48.20	27.84	29.11	30.42	53.85	65.28
MedRAG	Qwen	47.35	28.58	41.77	25.24	53.85	63.42
MedGENIE	Llama	48.02	29.94	18.99	21.42	**61.54**	65.15
TC-RAG	Qwen	49.43	31.44	29.11	28.92	30.77	66.17
CF-MoA (B)	Llama	58.29	44.63	48.10	53.21	53.85	68.84
CF-MoA (B)	Qwen	**59.36**	**46.62**	**58.23**	**60.03**	23.08	68.79

Each configuration covers 13,905 unique inputs. ALL averages non-reference task scores within units and then across units; R labels may overlap. Llama denotes Llama-3.1-8B-Instruct and Qwen denotes Qwen3-8B. The selected MedRAG-Llama baseline includes its output repair.

correctly while struggling to revise the support for a specific decision.

The taxonomy links these performance differences to specific reasoning operations. Support removal calls for checking whether an answer's prerequisites still hold; diagnosis calls for comparing viable candidates under the changed evidence. These requirements motivate the support-update and candidate-comparison heads evaluated next.

6.2 Specialized heads transfer across seven configurations

Figure 2 shows improvements in all 21 system–category comparisons. Support addition rises from an original range of 26.73–31.44% to 42.74–47.70%; support removal rises from 7.59–41.77% to 29.11–58.23%; and candidate comparison rises from 8.59–30.42% to 32.47–60.44%. The gains occur across direct answering, retrieval-based systems, and systems with their own agent or control loops. They show that a functional head can repair failures left by different upstream workflows.

Shared operations address the overlap between categories. On support addition, the adopted wrapper routes 1,780 units to diagnosis, 72 to rule execution, and retains the original result on 1,014 units. On candidate comparison, 680 units use diagnosis, ten use rules, and 43 retain their original result. A diagnosis decision or support update can therefore serve multiple functional categories within the same implementation.

Candidate presentation accounts for much of the diagnosis gain. On the 680-unit diagnostic subset, averaging across the seven systems, the original score is 20.97%, a single catalog decision reaches 45.44%, six uncalibrated decisions reach 46.47%, and calibrated decisions reach 46.13%. The largest common improvement therefore comes from asking the model to make an explicit decision over the available diagnoses. Additional voting and calibration have smaller, configuration-dependent effects. For MedRAG-Llama, the correct count changes from 203 to 339 with one catalog decision and to 359 with either six-decision variant. For MedRAG-Qwen, the corresponding counts are 160, 382, 402, and 410. Appendix B gives every configuration and additional source checks.

Table 2: Support-operation ablation on all 123 rule inputs, reusing the same extracted facts and execution states. Parentheses give the change in correct outputs relative to the complete kernel.

Policy	Llama	Qwen
Full support kernel	69/123	71/123
Without add	43/123 (-26)	48/123 (-23)
Without remove	53/123 (-16)	60/123 (-11)

A1 and A2 share one rule kernel. Unavailable outputs remain in the denominator.

Both support operations contribute. The shared support executor uses the extracted facts to add newly justified options and remove contradicted ones. On the 123-input control in Table 2, disabling addition reduces correct answers from 69 to 43 for Llama and from 71 to 48 for Qwen. Disabling removal reduces them to 53 and 60. Both operations contribute under the same fact extraction and rule statuses.

The support-removal category has a strong answer prior: a coverage-matched constant "None" control obtains more correct answers than the head in each of the seven configurations. On the nine covered prescribing examples with remaining support, the constant control scores zero; the head improves over the original answer in five configurations, is unchanged in one, and declines in one. Together with the operation ablation, these results locate the value of support execution in recomputing which options remain justified. The category average also reflects how often the correct answer is "None".

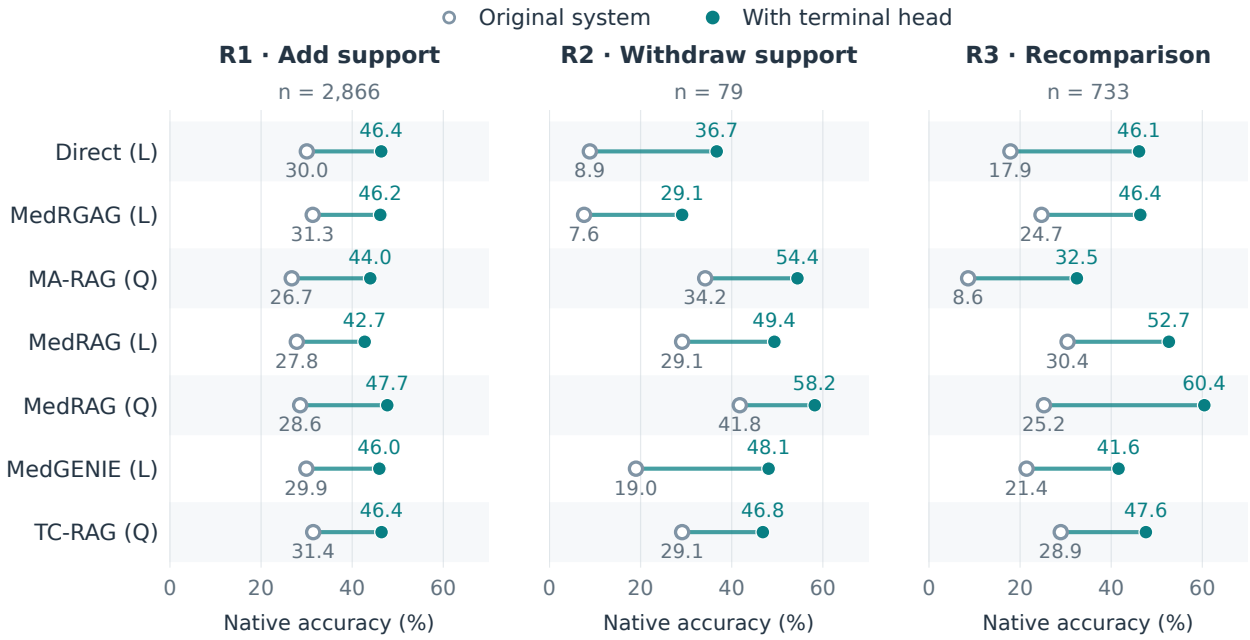


Figure 2: Original and head-augmented performance across all seven configurations. Each panel uses the complete corresponding category: support addition (n = 2,866), support removal (n = 79), and candidate comparison (n = 733). All 21 comparisons improve. Category memberships overlap; exact correct counts and branch coverage are in Appendix B.

6.3 The complete controller improves overall and paired correctness

On the complete benchmark, the aggregate score rises from 48.20 to 58.29 for MedRAG-Llama and from 47.35 to 59.36 for MedRAG-Qwen (Table 1). These within-configuration gains are 10.09 percentage points (95% family bootstrap interval: 8.96–11.23) and 12.01 points (10.92–13.15). Both configurations improve on support addition, support removal, candidate comparison, and preservation, and their final aggregate scores exceed the highest baseline score.

On the 13 clinical consequence units, MedRAG-Llama remains at 53.85 while MedRAG-Qwen decreases from 53.85 to 23.08 under the readout branch.

The controller improves both correct preservation and correct response to changed evidence (Figure 3). On answer-preserving pairs, both-side correctness rises from 58.57 to 64.93% for Llama and from 51.59 to 65.22% for Qwen. On answer-changing pairs, it rises from 0.73 to 19.49% and from 1.52 to 20.22%, respectively. All four paired differences have positive family-bootstrap intervals. The remaining gap between preserving and changing pairs shows that evidence-sensitive updating is the more difficult behavior in this evaluation.

6.4 Conditional execution and stable readout

The consequence agent addresses tasks with an explicit executable model. On 24 virtual physiological profiles with 12 questions each, typed local intervention produces correct binary and numerical outputs on all 288 questions for both backbones. A same-tools alternative that asks the model to generate a complete plan obtains 270 and 222 correct numerical outputs. Removing the intervention from the patch procedure reduces numerical correctness to 67 for both backbones. These controls support the value of a constrained local edit followed by execution in the supplied model (Appendix B.4).

Stable readout is evaluated on a separate 208-group component panel combining answer sampling and NLI interfaces. Its group-weighted paired score improves from 60.19 to 65.90% for Llama and from 59.17 to 66.47% for Qwen. In the final 16-group extension, both-side correctness improves from 11 to 12 groups and from 12 to 13 groups relative to the matched single-answer control. These observations motivated its use as the broad readout branch of the adopted controller.

6.5 Aggregation gains depend on the backbone

Further aggregation produces mixed gains on the same full input set. A rationale-conditioned verifier scores the

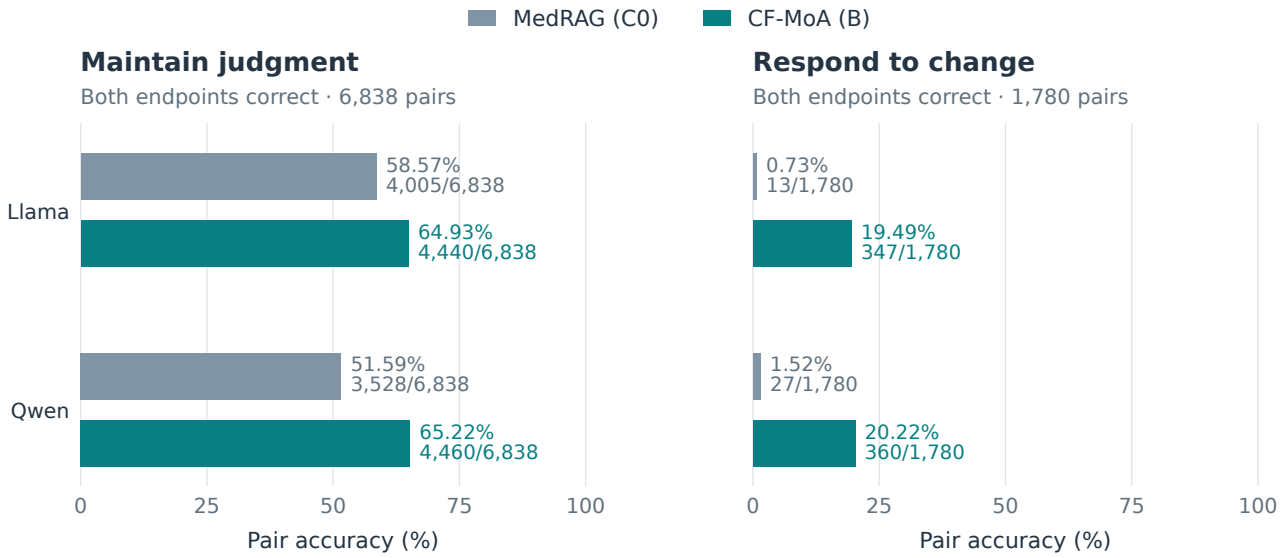


Figure 3: Correctness on both members of a pair. On answer-preserving pairs, correct counts increase from 4,005 to 4,440 for Llama and from 3,528 to 4,460 for Qwen ($n = 6{,}838$). On answer-changing pairs, they increase from 13 to 347 and from 27 to 360 ($n = 1{,}780$). Each pair counts as correct only when both answers match their respective references.

distinct candidates already present in the saved readout pool and returns an existing complete answer. Relative to B, it changes the aggregate score by -1.81 points for Llama and $+0.93$ points for Qwen. Its candidate-only and joint-selection counterparts score below B on both backbones. Three additional sampled answers yield 58.32 and 59.67, with more unavailable outputs. Appendix C reports the complete six-arm comparison. Earlier generated-answer aggregation also underperformed the adopted controller on its development panel. Across these variants, adding an aggregation stage does not consistently improve on sparse execution of the selected operations.

The adopted controller uses 59,800 attributed component requests for Llama and 58,301 for Qwen, in addition to original answering and retrieval costs. Rationale-conditioned verification adds 13,308 and 9,820 requests for the respective score changes above. Its added cost and backbone-dependent gains support retaining B as the default CF-MoA configuration. Appendix C.4 gives the physical request and token accounting.

7 Discussion and Conclusion

The five-relation taxonomy connects behavioral evaluation to the design of executable operations. Its category profiles reveal weaknesses hidden by aggregate rankings, and the head comparisons show that several of these weaknesses recur across upstream systems. CF-MoA organizes the resulting operations into an inference procedure that improves aggregate accuracy and correctness across related inputs.

The component controls suggest a practical design principle: make the required decision explicit. A catalog presents competing diagnoses for direct comparison. A support executor converts extracted facts into option-level checks with a defined treatment of unknown values. An executable model turns an intervention specification into a verifiable calculation. Each operation gives the system a concrete procedure for the behavior requested by the task. The large gain from a single catalog decision and the support-operation ablations show why these interfaces deserve attention alongside retrieval and answer generation.

CF-MoA assigns each agent an operational responsibility and routes inputs to the applicable computation. Agents share weights and execution where appropriate: a rule question uses support recomputation, while a diagnostic question uses candidate comparison. Further verification adds requests and produces different aggregate effects across the two backbones. This tradeoff favors the adopted sparse controller for the evaluated configurations and makes candidate selection a separate empirical design choice.

The 11-source benchmark, seven-configuration diagnosis, 21 head comparisons, and two full-system evaluations establish a connected path from functional analysis to system design. The resulting framework measures two complementary abilities: changing a decision when its evidence changes and preserving it when its support remains intact. Extending these operations to new case families and resource settings provides a con-

crete direction for further counterfactual medical reasoning research.

Limitations

The benchmark and component panels informed method development. Bootstrap intervals quantify uncertainty within these evaluated families; performance on unseen families requires a separate family-disjoint evaluation after final controller selection. System coverage comprises two MedRAG configurations for the final controller and seven configurations for the earlier support and comparison heads.

Category coverage is uneven: support removal contains 79 units and consequence calculation contains 13. Labels overlap, and a substantial part of the preservation subset comes from perturbation and retrieval-robustness tasks. Taxonomy assignments underwent structured review and source-specific checks; independent clinician agreement remains unmeasured. Reference labels and scoring rubrics inherit the properties of their source datasets.

Component performance depends on the supplied resources. The rule executor uses a 71-rule bank and extracted facts, diagnosis uses a finite public catalog, and the consequence agent was validated on a supplied physiological model and virtual profiles. The measured gains include contributions from candidate coverage and readable output. Isolating changes in latent clinical reasoning requires controlling these resources. Retrieval, formatting, and inference budgets differ across baselines; a complete resource-matched comparison remains open.

Ethical Considerations

The study evaluates existing research datasets and simulated profiles. Patient-care use requires clinical validation beyond these evaluations. Dataset provenance and original access conditions must accompany redistribution. Evaluation artifacts include source task contracts, failures, and invalid outputs for inspecting behavior at the individual-input level. Author and data-availability declarations appear in Appendix E.

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Table 3: Complete v5 benchmark composition by source. R1–R5 columns count units bearing each category label.

ID	Source	Units	Judgments	R1	R2	R3	R4	R5
M01	MedEinst	1,780	2,816	1,780	0	680	0	0
M02	MedPIC-Bench	167	182	96	75	10	0	0
M06	HIV causal scenarios	13	13	0	0	0	13	0
M10	CPV-MedQA	451	451	1	0	0	0	450
M11	Counterfactual Cultural Cues	517	517	67	0	0	0	450
M12	FairMedQA/AMQA	584	584	134	0	43	0	450
M14	DiversityMedQA	454	454	0	4	0	0	450
M17	MedPerturb	450	1,344	0	0	0	0	450
M20	MedCounterFact	788	1,720	788	0	0	0	450
M22	MedRGB	450	4,469	0	0	0	0	450
M23	BioRAB	450	450	0	0	0	0	450
Total		6,104	13,000	2,866	79	733	13	3,600

The benchmark contains 6,104 unique units and 13,000 target judgments. Category memberships overlap and must not be summed as independent units. Target judgments are distinct from the 13,905 inference inputs and 15,616 native scoring mappings per system. The separate supplied-model A4 panel is not added to this table.

A Benchmark Composition and Annotation

A.1 Source inventory and qualification

Table 3 gives the exact composition of the frozen evaluation. Source citations and acquisition records accompany the source manifest. The original inventory contained 26 candidates, of which 15 were acquired. Four acquired resources did not contribute eligible v5 targets: the clinical laboratory and EquityMedQA resources required additional answer or rubric adjudication; MedCF required an editing operation outside the implemented answering protocol; and MedMKEB lacked the required images. For the other 11 candidates, the specified fixed cases, counterfactual edits, or required restricted context had not been obtained at collection time.

The 55,991 archived units comprise 53,494 text units and 2,497 metadata units with missing images. An initial review frame contained 28,243 units and 80,910 judgments. Its review ledger records 603 individual semantic reviews, 13,464 grouped semantic reviews, and 14,176 structural checks. Subsequent completion and eligibility review covered all archived units, producing 109,599 reviewed judgments, of which 77,781 were eligible in 38,123 units. The evaluation sample is drawn from this completed pool.

A.2 Annotation examples and overlap

An annotation identifies the source comparison axis before assigning a relation. In a MedEinst case, the addition of wheezing supports a bronchitis candidate (R1), while the associated diagnostic question can require comparison with acute laryngitis (R3). A DiversityMedQA sex-attribute change removes one demographic basis for a Lesch–Nyhan candidate (R2), while other findings remain available. In CPV-MedQA, changing an ethnicity description leaves the chemotherapy and ototoxicity evidence relevant to a pharmacological question intact (R5). Each label describes a relation between a specified change and its target in the full case context.

Labels are assigned to targets and then combined at the unit level. In v5, R3 is contained in R1; R1 and R5 share 450 units, R1 and R2 share four, and R2 and R3 share three. R1–R3 contain 2,941 distinct units. The sum of category memberships is 7,291, whereas the union used for ALL contains 6,104 units. The 13,000 selected target judgments, 13,905 distinct inputs, and 15,616 scoring mappings refer to different representations of these same tasks. Of the scoring mappings, 6,818 address reference endpoints and 8,798 address non-reference endpoints.

A.3 Sampling and version lineage

The earlier v2 evaluation contained 3,105 units and eight sources. A v3 display assigned overlapping units to a single category for presentation, preserving their answers. The completed v4 pool restored multi-label evaluation

and expanded eligibility review. The v5 evaluation used in this paper retains all eligible R1–R4 targets and samples 450 R5 units from each of eight sources. These source samples are drawn without replacement from sorted identifiers using a source-specific seeded procedure. The original semantic labels and the selected evaluation labels are stored separately. Eligibility and seeded sampling determine membership independently of predictions and method scores.

The 450-unit allocation balances source representation within R5, with sampling fractions defined by each source's eligible pool. Related variants remain linked to their parent families for paired analysis and bootstrap resampling. The source manifest, membership files, and scoring mappings define the evaluated sample.

B Component Evidence and Controls

B.1 Complete head comparisons

Table 4 reports correct counts for every configuration in the head comparison. The R1 wrapper uses diagnosis on 1,780 units, support rules on 72, and the saved original answer on 1,014. R2 uses rules on 55 units and preserves 24 original answers. R3 uses diagnosis on 680 units, rules on ten, and preserves 43 original answers. These are unit-level coverage counts for the head experiment; the full controller's input-level coverage additionally includes reference endpoints and other compatible tasks.

B.2 Diagnosis: catalog, calibration, and source checks

The public catalog contains 49 diagnoses, while the frozen task scorer recognizes 46 normalized diagnosis keys. The fixed public catalog is supplied to the decision component independently of the current item's gold answer. On 680 diagnostic units, Table 5(a) separates the single-catalog improvement from repeated selection and calibration. Across repaired R3 outcomes, 1,186 of 1,272 previously failed outputs were unmapped diagnoses, 23 were otherwise invalid, and 63 were valid wrong answers. This breakdown indicates that output grounding accounts for a large part of the measured improvement. The task scorer treats unmapped outputs as incorrect.

Additional controls explored alternative name/code presentations on 40 diagnostic inputs. Direct code selection with a 384-token budget obtained 18/21 correct for Llama/Qwen; direct name selection at a 128-token budget obtained 15/18. Reasoning with names at 384 tokens obtained 18/19, while reasoning with codes at that budget obtained 21/22. The direct-name/direct-code comparison jointly varies representation and token budget. The evidence ledger retains the complete run identities and budgets.

Two additional source checks assessed catalog compatibility. On 47 ASCENT records, the calibrated head obtained 38/39 correct for Llama/Qwen, compared with 39/39 for uncalibrated voting and 38/39 for a single catalog decision. These records included 42 source-training and five source-test examples and subsequently informed development. Across three CUPCase batches totaling 18 records, the calibrated head obtained 15/16, uncalibrated voting 16/17, single selection 15/18, and character-length-normalized candidate likelihood 15/14. Catalog-based selection performs well on these additional records; calibration provides no consistent incremental advantage.

B.3 Support removal and remaining support

The constant "None" control in Table 5(b) uses the same 55-input rule coverage and retains the same 24 original answers outside coverage. Forty-six covered targets have "None" as their answer; the other nine retain support for at least one option. On those nine, original correct counts for Direct, MedRGAG, MA-RAG, MedRAG-Llama, MedRAG-Qwen, MedGENIE, and TC-RAG are 5/4/1/3/0/2/3, and head counts are 6/4/3/4/6/5/2. This subset measures the ability to retain independent support after another basis is removed.

Table 4: Correct counts before and after the specialized heads. Category memberships overlap; R1–R3 contain 2,941 distinct units. Section 5 specifies context reuse and rule-fact sharing.

System	R1 ($n = 2{,}866$)	R2 ($n = 79$)	R3 ($n = 733$)
M0 Direct	$861 \rightarrow 1,329$	$7 \rightarrow 29$	$131 \rightarrow 338$
M2 MedRGAG	$898 \rightarrow 1,323$	$6 \rightarrow 23$	$181 \rightarrow 340$
M3 MA-RAG	$766 \rightarrow 1,260$	$27 \rightarrow 43$	$63 \rightarrow 238$
M4 MedRAG–L	$798 \rightarrow 1,225$	$23 \rightarrow 39$	$223 \rightarrow 386$
M5 MedRAG–Q	$819 \rightarrow 1,367$	$33 \rightarrow 46$	$185 \rightarrow 443$
M6 MedGENIE	$858 \rightarrow 1,317$	$15 \rightarrow 38$	$157 \rightarrow 305$
M7 TC-RAG	$901 \rightarrow 1,331$	$23 \rightarrow 37$	$212 \rightarrow 349$

Table 5: Diagnosis and support-removal controls. (a) Original answers, single catalog selection, six-order voting, and calibrated voting on 680 diagnostic units. (b) Head and coverage-matched constant "None" results on R2, with the nine covered cases retaining support shown separately.

(a) R3: correct out of 680

System	Original	Order 0	Majority	Calibrated
M0 Direct	116	299	314	314
M2 MedRGAG	163	320	325	315
M3 MA-RAG	33	215	200	204
M4 MedRAG–L	203	339	359	359
M5 MedRAG–Q	160	382	402	410
M6 MedGENIE	137	296	288	276
M7 TC-RAG	186	312	324	318

(b) R2: aggregate and remaining-support subset

System	Head /79	None /79	Head /9	None /9
M0 Direct	29	48	6	0
M2 MedRGAG	23	47	4	0
M3 MA-RAG	43	58	3	0
M4 MedRAG–L	39	58	4	0
M5 MedRAG–Q	46	56	6	0
M6 MedGENIE	38	52	5	0
M7 TC-RAG	37	54	2	0

The 123-input add/remove ablation in the main text holds the extracted facts and statuses fixed. It evaluates the shared executor on the controller's complete rule-input panel, which extends beyond the 79-unit R2 category. The full executor leaves one Llama input and 14 Qwen inputs unavailable; these remain included in the 123-input denominator. The contrast tests the effect of the specified operations under the same upstream extraction.

B.4 Supplied-model consequence execution

The separate model panel uses 24 virtual parameter and initial-state profiles, each with 12 questions. A question supplies the physiological mechanism, factual schedule, requested local intervention, and output projection. The typed-patch procedure edits the named action and executes the resulting trajectory. The full-plan control has the same tool access but regenerates a complete plan; the no-intervention control retains patch structure while omitting the actual intervention. Table 6 reports binary, numerical, and full-output correctness for the 288 structured queries over these virtual profiles.

Table 6: Supplied-model consequence execution. Each configuration contains 288 questions from 24 virtual parameter/initial-state profiles. Counts report correct outputs within the supplied model contract.

Config.	Procedure	Binary /288	Numerical /288	Full output /288
M4	Ordinary reanswer	118	0	0
M4	Full execution	286	270	270
M4	Patch without intervention execution	234	67	67
M4	Typed patch and execution	288	288	288
M5	Ordinary reanswer	187	0	0
M5	Full execution	265	222	222
M5	Patch without intervention execution	234	67	67
M5	Typed patch and execution	288	288	288

B.5 Stable readout and generated aggregation

The F component panel comprises an initial 64 groups, an additional 128 groups, and a final frozen extension of 16 groups. Group-weighted paired scores for the complete 208-group panel are reported in Section 6.4. The ordinary branch aggregates valid native answer keys; the NLI branch combines direct and analysis-conditioned semantic

probabilities across both label mappings. The component's exact stopping, tie-breaking, and invalid-output policies are specified in Section 4 and the accompanying implementation.

The development study also tested generated aggregation over the available expert outputs on 121 inputs. The adopted controller obtained 87/82 correct for Llama/Qwen, a three-draw alternative 84/74, and the default generated aggregator 75/73. Adding a newly developed support-addition proposal to the candidate pool produced 72/57 under top-one selection and 79/74 under generated aggregation. These experiments led to the shared support kernel and sparse adopted controller. The 121-input panel serves as a development control separate from the full-v5 evaluation.

C Complete System Results and Aggregation Variants

C.1 Six-arm comparison

The full evaluation uses the same input and scoring-mapping sets for all six arms. Table 7 defines the arms, and Table 8 reports aggregate scores, correct-input counts, and unavailable outputs. Correct-input counts use input-level scoring and are therefore distinct from the unit-macro ALL score. Table 9 retains all five functional categories for every arm; Table 10 gives source-specific C0/B results, and Table 11 gives their native input-format results.

Table 7: Six evaluation arms. C1 and B denote the same adopted operation core.

Arm	Decision rule	Role in the comparison
C0	Original MedRAG configuration	Original-system reference
C1 / B	Adopted operation core	Adopted CF-MoA controller
C2	B plus candidate-only binary verification	Same-pool verification without saved rationales
C3	B plus rationale-conditioned binary verification	Same-pool verification with saved rationales
C4	B with old F votes plus three additional answers	Expanded-pool sampling control
C5	B plus joint selection over the same candidates and rationales	Alternative same-pool selection control

Table 8: Full-v5 results. Each row contains 13,905 inputs, 15,616 scoring mappings, and 6,104 non-reference native units. ALL is the mean of native-unit scores. Unavailable inputs remain in the denominators.

Config.	Arm	ALL (%)	Correct inputs	Unavailable inputs
M4 Llama	C0	48.1990	7,441	0
M4 Llama	B	58.2924	8,924	51
M4 Llama	C2	55.9939	8,554	51
M4 Llama	C3	56.4815	8,649	51
M4 Llama	C4	58.3219	8,951	157
M4 Llama	C5	56.7295	8,660	51
M5 Qwen	C0	47.3487	6,913	0
M5 Qwen	B	59.3616	8,971	22
M5 Qwen	C2	59.3562	8,843	22
M5 Qwen	C3	60.2889	9,060	22
M5 Qwen	C4	59.6723	8,878	86
M5 Qwen	C5	58.9690	8,739	22

C.2 Paired outcomes and uncertainty

Table 12 reports the complete both-endpoints-correct counts. All 1,780 answer-changing pairs come from MedEinst; the preserving-pair set spans multiple sources. The candidate-selection extensions operate on ordinary readout disagreements, so the diagnostic response-pair results remain identical across B and the four extensions in this evaluation. The family bootstrap resamples parent families as clusters and recomputes paired differences, preserving all units from a sampled family. Table 13 reports the principal aggregate contrasts, and the machine-readable result package contains all 36 recorded contrast intervals across aggregate and paired metrics.

Table 9: Full six-arm category scores (percent). ALL and category scores use native unit means; categories overlap.

Backbone	Arm	ALL	R1	R2	R3	R4	R5
Llama	C0	48.20	27.84	29.11	30.42	53.85	65.28
Llama	B	58.29	44.63	48.10	53.21	53.85	68.84
Llama	C2	55.99	43.79	48.10	52.11	46.15	65.61
Llama	C3	56.48	44.87	48.10	52.80	30.77	66.05
Llama	C4	58.32	44.87	48.10	52.52	38.46	69.11
Llama	C5	56.73	44.80	48.10	52.39	38.46	66.49
Qwen	C0	47.35	28.58	41.77	25.24	53.85	63.42
Qwen	B	59.36	46.62	58.23	60.03	23.08	68.79
Qwen	C2	59.36	48.43	58.23	60.03	30.77	68.39
Qwen	C3	60.29	49.02	59.49	60.30	30.77	69.70
Qwen	C4	59.67	48.15	58.23	60.30	38.46	68.93
Qwen	C5	58.97	47.87	58.23	59.89	30.77	67.99

Table 10: Source-specific native unit means (percent) for the original system and adopted controller. Source sizes are comparison units.

Source	Units	Llama C0	Llama B	Qwen C0	Qwen B
MedEinst	1,780	13.26	35.62	10.90	40.34
MedPIC-Bench	167	28.74	53.89	38.32	58.08
HIV scenarios	13	53.85	53.85	53.85	23.08
CPV-MedQA	451	68.74	69.62	65.85	65.85
Cultural Cues	517	67.31	70.02	68.67	70.60
FairMedQA/AMQA	584	61.47	66.78	64.04	63.01
DiversityMedQA	454	72.25	72.91	67.40	69.82
MedPerturb	450	87.48	91.15	76.26	81.52
MedCounterFact	788	51.02	55.33	56.22	53.05
MedRGB	450	69.20	72.00	52.89	74.13
BioRAB	450	44.22	57.78	59.78	75.56

Table 11: Native input-format accuracy (percent), including reference inputs. This input-level metric differs from ALL.

Format	Inputs	Llama C0	Llama B	Qwen C0	Qwen B
diagnosis	3,560	9.47	38.01	9.49	42.67
multi	167	28.74	53.89	38.32	58.08
relation	989	52.78	55.21	59.45	55.61
robustness_single	2,700	69.56	71.96	53.00	74.44
single	6,489	71.75	76.93	69.22	73.89

Table 12: Both-endpoints-correct counts for all six arms. Maintain uses 6,838 pairs and respond uses 1,780 pairs.

Arm	M4 maintain	M5 maintain	M4 respond	M5 respond
C0	4,005	3,528	13	27
B	4,440	4,460	347	360
C2	4,139	4,381	347	360
C3	4,144	4,486	347	360
C4	4,509	4,412	347	360
C5	4,150	4,246	347	360

Table 13: Paired contrasts in the ALL mean. Effects and 95% family bootstrap intervals are in percentage points. These are descriptive intervals on exposed development data.

Config.	Contrast	Effect	95% interval
M4	B − C0	+10.0934	[+8.9580, +11.2340]
M4	C3 − B	−1.8108	[−2.5287, −1.0819]
M4	C3 − C2	+0.4877	[−0.1275, +1.0907]
M5	B − C0	+12.0129	[+10.9177, +13.1510]
M5	C3 − B	+0.9273	[+0.3039, +1.5562]
M5	C3 − C2	+0.9327	[+0.4641, +1.4010]

C.3 Candidate verification and alternative selection

For an ordinary F input with at least two distinct valid answer keys, we retain the earliest complete saved response representing each key. C2 scores whether each candidate is correct under two opposite A/B label assignments. C3 uses the same procedure and includes the saved candidate rationale as model-generated reasoning to be checked. For candidate k, the verifier averages its normalized probability of correctness under the two assignments:

$$v(k) = \frac{1}{2}\left[p_{AB}(\text{correct} \mid x, k, r_k) + p_{BA}(\text{correct} \mid x, k, r_k)\right]. \tag{3}$$

The rationale r_k is omitted in C2. The highest-scoring candidate is selected; exact ties favor the saved F answer, then native order. The complete existing representative is returned, preserving its required auxiliary fields. Non-ordinary F inputs, single-candidate pools, and unavailable B outputs retain B. Technical verification failure also falls back to B and is recorded.

The full ledger identifies 2,964 Llama and 2,295 Qwen ordinary disagreement inputs. Two Llama B outputs are unavailable, leaving 2,962 actually verified Llama inputs. C3 records one Llama and eight Qwen technical verifier failures with fallback. C5 scores a joint finite choice over the same candidates and rationales. C4 adds three predetermined native-answer draws to the old F pool and votes over the expanded pool, retaining its own unavailable-output policy.

C.4 Computation and attribution

Table 14 reports the adopted component cost and the additional C3 cost. B uses 241,675,469 input and 9,686,224 output tokens for Llama, and 220,580,598 input and 7,645,205 output tokens for Qwen. Across the shared full-evaluation execution of all arms, the physical ledger contains 162,979 requests, 957,810,324 input tokens, and 18,291,906 output tokens. Method-attributed costs allocate the shared component once to each evaluated method; the physical ledger counts each executed request once.

Table 14: Attributed request costs for B and C3. The table includes the shared core once per method and excludes unknown common C0/retrieval costs. Tokens are the total attributed C3 tokens, not only verifier tokens.

Config.	B requests	C3 extra requests	C3 total requests	C3 input tokens	C3 output tokens
M4	59,800	13,308	73,108	358,643,985	9,699,532
M5	58,301	9,820	68,121	294,927,306	7,655,025

Original C0 answering and retrieval costs remain unknown because recording is incomplete. Timing records cover 11,402 of 13,905 component input rows per configuration, leaving 2,503 without complete timing. End-to-end runtime is unavailable. The physical request totals cover the later full-system experiment.

D Implementation and Reproducibility

D.1 Configuration identities

The selected seven-system comparison uses the following implementation identities: M0, direct Llama-3.1-8B-Instruct; M2, local all-Llama MedRGAG; M3, intrinsic MA-RAG with Qwen3-8B; M4, MedRAG-Llama with the selected structured-output repair; M5, MedRAG-Qwen; M6, MedGENIE v2 with Llama; and M7, TC-RAG with

Qwen and typed control v2. The paper uses system names, while manifests retain these identifiers. The discontinued i-MedRAG experiment is preserved separately under its own identity in the historical archive.

Selected outputs cover all 13,905 native inputs per configuration. Exact reuse is recorded for M0 (8,877 reused and 5,028 newly inferred), M2 (6,687 and 7,218), and M3 (3,690 and 10,215). M4–M7 each have 13,905 newly inferred outputs in their selected runs. Reuse requires the same input, method configuration, and output contract. Each selected method has a fixed output contract that includes its structured-output handling; earlier runs are archived separately.

D.2 Rebuilding paper artifacts

The paper package separates native LaTeX source, frozen numeric inputs, plotting scripts, and audit receipts. The main manuscript is maintained in `latex/main.tex`, with section files, tables, vector figures, and `references.bib`. Running `latex/build.sh` compiles it using Tectonic or a local LaTeX toolchain.

The reproduction scripts read frozen summaries and regenerate the main tables, source and format breakdowns, and vector figures. They retain source hashes and assert denominators and selected values before exporting results. The source package includes all assets required to compile the paper offline once the TeX dependencies are installed. Its companion evidence package identifies the original configuration, prediction, score, and trace artifacts for each experiment. Dataset licenses and large model/retrieval assets remain governed by their original distribution conditions.

The deterministic artifact rebuild reproduces the displayed results from saved experiment outcomes. Re-running inference additionally requires the exact backbone checkpoints, component prompts, rules and catalogs, native-input manifests, retrieval assets, and component-specific runtime settings. The evidence index lists these inference dependencies separately from the artifact-build requirements.

D.3 Evidence organization

The data construction record defines eligibility and version lineage; the evaluation protocol defines native scoring and input permissions; the selected-system record defines the seven baseline versions. Head-specific records contain their implemented rules, control experiments, and transfer results. The full-system closeout binds the six arms to source/category/format results, paired intervals, and request costs. A separate coauthor dossier connects these records chronologically. The manuscript draws its claims from this evidence map while retaining the original run identities and unsuccessful variants in the companion archive.

E Author and Availability Declarations

Data and code availability. The manuscript source, figure-generation scripts, frozen result summaries, and experiment evidence indexes are maintained in the authors' private repository. Redistribution of source datasets must follow each original provider's terms; where redistribution is restricted, a release can provide identifiers, acquisition instructions, and derived membership or scoring metadata.

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AI assistance. An artificial intelligence assistant was used to organize existing experiment records, draft and edit text, prepare LaTeX, generate plotting code, and check consistency against saved artifacts. All reported evaluation results, labels, and predictions come from saved experiment artifacts. The authors remain responsible for verifying the manuscript, disclosures, citations, and source-data permissions.